

2 (a) Cholera, malaria and tuberculosis are infectious diseases. Vaccines are available for these three diseases.

(i) State why these three diseases are described as infectious diseases.

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..... [2]

(ii) There is no vaccine to provide protection from infection by HIV.

Suggest why it has been difficult to develop a vaccine to provide protection from infection by HIV.

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..... [2]

(b) Figure 2.1 shows how the number of new cases of HIV across the world changed between 1990 and 2022.

In 2015, the United Nations set a target to reduce the number of new cases of HIV as part of the programme to control HIV infection. This target is 0.18 million by 2030.

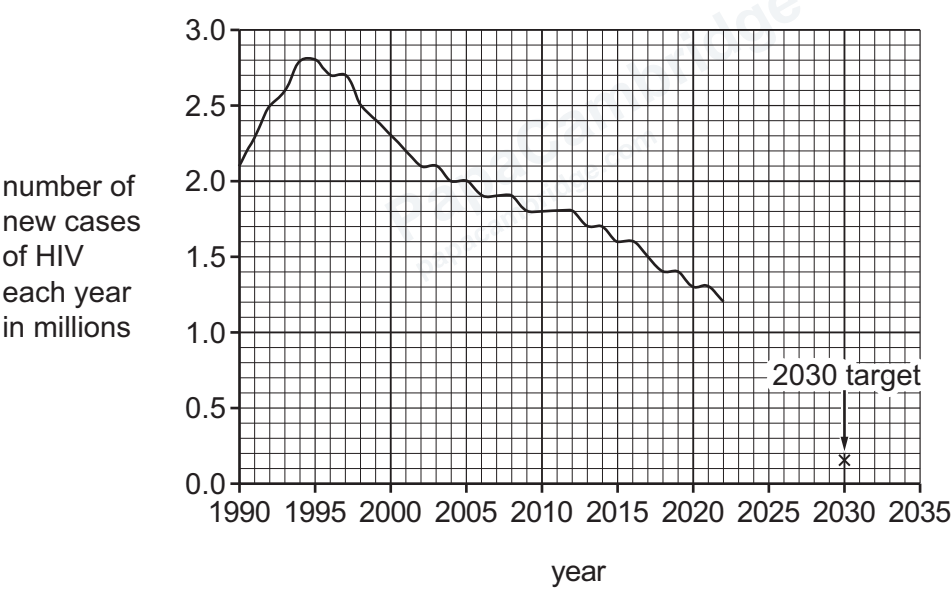


Figure 2.1

(i) Calculate the percentage decrease in the number of new cases of HIV from 2022 that will be needed to reach the target for 2030.

Space for working.

.....% [1]

(ii) Table 2.1 lists two of the aims set by health authorities to reduce the number of new cases of HIV.

Complete Table 2.1 by suggesting ways in which health authorities can achieve these aims.

Table 2.1

aim	way to achieve aim
to reduce the risk of HIV infection during blood transfusions	
to reduce the risk of new-born babies being infected with HIV	

[2]

[Total: 7]